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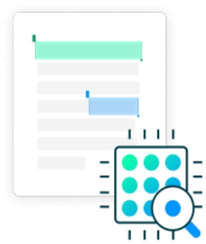

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Intelli Gene: Smart Assessment of Hereditary Disorders Risk

Sudha V
Department of Computer Science and Engineering,
Panimalar Engineering College,
Chennai, India
sudhagandhi611994@gmail.com

Shunmathi S V
Department of Computer Science and Engineering,
Panimalar Engineering College,
Chennai, India
svshunmathi@gmail.com

Aparna Devi S
Department of Computer Science and Engineering,
Panimalar Engineering College,
Chennai, India
aparnadevi0704@gmail.com

Abstract **- Intricate connections between maternal health conditions, physiological factors, and genetic predisposition are the root cause of genetic and inherited disorders. Early medical intervention and preventive measures are made possible by the prompt identification of hereditary risk. Conventional diagnostic methods primarily rely on the expertise of clinicians and sequential investigative techniques, which can be labor-intensive, inaccurate, and impractical for screening at the population level. In order to address these issues, this study presents Intelligene, a web-based intelligent screening platform that uses ensemble-based machine learning models to predict hereditary disorder susceptibility early. To provide real-time risk estimation, the suggested framework integrates physiological measurements, maternal health parameters, family history indicators, and clinical characteristics. To improve model performance and reliability, a methodical data pre-processing workflow is used, which includes noise reduction, categorical encoding, feature normalization, and feature extraction. To increase classification stability and overall accuracy, the predictive architecture uses an ensemble learning strategy that combines Random Forest, Bagging, and Naive Bayes classifiers. Unlike traditional chromosomal or genomic testing methods, Intelligene only uses routinely collected clinical data, which drastically lowers costs and complexity without sacrificing predictive power. Prioritized screening and well-informed clinical decision-making are supported by the system's consistent and accurate risk predictions, as confirmed by performance evaluation. For early-stage hereditary disorder assessment, the suggested solution is ideally suited for implementation in actual healthcare settings due to its strong scalability and practicality.**

Keywords—Machine Learning, Genetic Disorder Prediction, Ensemble Learning, Healthcare Analytics, Risk Assessment, Web-based Screening System

I. INTRODUCTION

Genetic and heritable diseases are a major worldwide health concern, and they lead to higher morbidity, long-term incapacity, as well as reduced quality of life. These conditions result from intricate interactions between inherited genetic susceptibilities, maternal health, environmental insults and biologic processes. Early recognition of those at increased hereditary risk allows for timely clinical management and personalized treatment planning, as well as risk-reduction strategies, with the ultimate goal of reducing morbidity and mortality through prevention or early treatment.

Traditional diagnostic approaches for genetic diseases are predominantly based on the clinical acumen of a specialist combined with a comprehensive physical evaluation followed by a series of tests specific to that disorder. These approaches are highly successful in clinical settings where the subject is examined under relatively controlled conditions, although they can be labour-intensive and time-consuming and become limited by inter-observer variability. Also, population level screening based on conventional diagnostic platforms are challenging due to high operational costs, lack of accessibility and necessary specialized laboratory facilities. These limitations make it challenging to detect early risks in low-resource healthcare environments.

Data-driven methods have become viable substitutes for clinical decision support and intelligent disease risk prediction due to the quick expansion of digital healthcare systems and developments in machine learning. Automated screening and early disease detection are made possible by machine learning techniques, which allow the extraction of intricate patterns from high-dimensional healthcare data. Using genomic and chromosomal datasets, numerous studies have shown how predictive modeling works well for analyzing genetic disorders. Nevertheless, these methods frequently necessitate expensive genetic testing, advanced lab apparatus, and professional interpretation, which limits their usefulness in standard clinical procedures.

This paper suggests Intelli Gene, a web-based intelligent screening framework for early hereditary disorder susceptibility prediction using ensemble machine learning models, as a solution to these problems. To produce precise and up-to-date risk assessments, the suggested system combines routinely gathered clinical characteristics, family inheritance markers, maternal health metrics, and physiological measurements. The framework improves predictive accuracy, robustness, and generalization capability by utilizing an ensemble learning strategy that combines Random Forest, Bagging, and Naive Bayes classifiers. This eliminates the need for costly laboratory-based genetic testing procedures.

II. LITERATURE REVIEW

Machine learning techniques have increasingly been applied to genetic and hereditary disease prediction due to their ability to model complex, high-dimensional biomedical data. Several studies have explored supervised learning models for identifying disease susceptibility using genomic and clinical features. Bracher-Smith et al. [1] conducted a systematic review on machine learning approaches for genetic risk prediction and highlighted the effectiveness of supervised classifiers in modeling polygenic disorders. However, most of the reviewed approaches primarily relied on genomic markers, limiting their practicality for routine clinical screening.

Regularized and ensemble-based learning methods have also been investigated for improving predictive robustness in healthcare analytics. Okser et al. [2] demonstrated that machine learning models can enhance risk prediction for complex traits when compared to traditional statistical approaches. Nevertheless, these models largely focused on genotype-level data and required access to detailed molecular information, increasing computational and infrastructural demands.

Feature selection and dimensionality reduction have been recognized as essential components in precision medicine applications. Pudjihartono et al. [3] emphasized that appropriate feature engineering significantly influences classification performance in biomedical datasets. While their study provided methodological insights, it did not specifically address hereditary disorder family-level stratification or real-time deployment considerations.

Several researchers have examined the use of ensemble learning for healthcare risk prediction. Verma and Gupta [5] reviewed ensemble techniques for medical diagnosis and concluded that combining multiple classifiers improves stability and generalization. Similarly, Li et al. [6] applied ensemble machine learning models for early disease risk screening and reported improved accuracy over individual classifiers. However, many of these frameworks were developed for general disease prediction and did not focus specifically on inheritance-aware risk categorization.

More recent studies have explored non-invasive and clinical-data-driven prediction approaches to reduce dependency on genomic testing. Banerjee and Ghosh [11] investigated clinical data mining methods for genetic risk analysis without direct genomic sequencing. While this approach improves accessibility, it does not incorporate multi-level disorder classification or structured follow-up prioritization.

Overall, existing research either concentrates on genomic-based disease prediction, focuses on single-disease diagnosis, or lacks integrated real-time screening with disorder-level classification. These limitations motivate the development of the proposed Intelli Gene framework, which integrates ensemble learning with routinely collected clinical, maternal, and inheritance-related features to enable scalable and practical hereditary risk assessment.

III. PROPOSED METHODOLOGY

The proposed Intelli Gene framework is designed as an end-to-end intelligent screening system for hereditary disorder risk assessment using structured clinical and inheritance-based data. The methodology follows a systematic pipeline comprising data acquisition, data processing, ensemble machine learning inference, and result generation. The overall workflow of the system is illustrated in the system architecture diagram (Fig. 1).

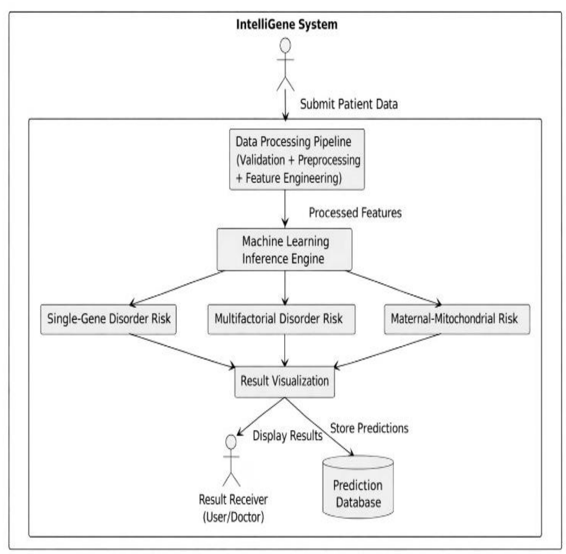

Fig. 1. Architecture Diagram of the Proposed Intelli Gene System

A. Data Collection and Feature Selection

The dataset used in this study consists of routinely collected clinical and inheritance-related information. Instead of relying on complex genomic sequencing data, the model utilizes practical indicators that are commonly available in standard healthcare settings. These include physiological measurements (such as blood parameters and vital signs), maternal health history, paternal inheritance details, demographic information, and selected pregnancy-related factors.

The selected features were chosen to capture both hereditary influence and non-genetic risk contributors. By combining maternal and paternal inheritance indicators with clinical observations, the framework is designed to reflect real-world screening conditions. This approach improves accessibility and reduces dependency on expensive chromosomal or DNA-based diagnostic procedures.

The age distribution of individuals included in the dataset is illustrated in **Fig. 2**, highlighting the demographic representation considered during model training and evaluation.

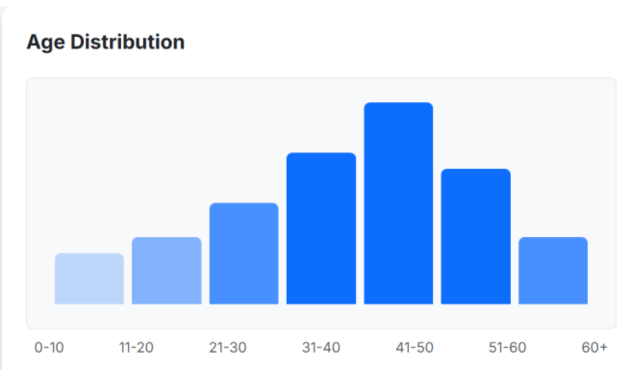

Fig. 2. **Age distribution of patients in the dataset**

B. Data Processing Pipeline

A unified data processing pipeline comprising input validation, data cleaning, categorical encoding, normalization, and feature engineering is applied to the raw dataset prior to model training. This pipeline enhances model generalization, lowers noise, and guarantees data consistency. To keep only pertinent characteristics that have a major impact on the classification of hereditary disorders, feature engineering techniques are used.

C. Ensemble Machine Learning Model

An ensemble learning approach is used to increase prediction robustness and reliability. The Random Forest, Bagging, and Naive Bayes classifiers are all integrated into the framework. Because Random Forest can model intricate nonlinear relationships between clinical and inheritance features, it is the main learner. In order to lessen variance and counteract overfitting, which is prevalent in medical datasets, bagging is used. As a probabilistic baseline classifier, Naive Bayes facilitates quick inference and comparative analysis.

D. Disorder Risk Classification

Single-gene disorder tendency, multifactorial disorder tendency, and maternal-linked mitochondrial disorder tendency are the three hereditary disorder risk categories into which the trained ensemble model divides the input data. The system predicts follow-up urgency levels in addition to classifying disorders.

E. System Deployment and Inference

In order to facilitate real-time inference and user interaction, the trained model is implemented as an online application. Predictions are produced instantly upon data submission and shown via a result visualization module.

The system database safely stores all input data and predictions for later review and analysis. Scalability, accessibility, and uniform screening performance for numerous users are guaranteed by the modular architecture..

IV. RESULTS AND DISCUSSION

Standard classification metrics, such as accuracy, precision, recall, and F1-score, were used to assess the performance of the suggested Intelli Gene framework. Both follow-up urgency prediction and hereditary disorder risk classification were evaluated experimentally. The outcomes show that the suggested ensemble-based method produces consistent and trustworthy prediction performance.

A. Evaluation of Model Performance

The accuracy comparison between the suggested ensemble model and individual machine learning models is shown in Fig. 2. Due to their capacity to identify intricate relationships in clinical and inheritance-based data, Random Forest and Bagging classifiers outperformed Naive Bayes in terms of accuracy. The highest accuracy was attained by the suggested ensemble model, suggesting that combining several classifiers improves generalization and robustness.

B. Category-wise Disorder Risk Analysis

A class-wise analysis was carried out to assess how well the suggested Intelli Gene framework performed across various hereditary disorder categories.

Disorder Category	Precision	Recall	F1-Score
Single-Gene Disorder	0.82	0.77	0.79
Multifactorial Disorder	0.88	0.86	0.87
Maternal-Mitochondrial Disorder	0.84	0.80	0.82

Table I. Class-wise performance evaluation of hereditary disorder categories

The suggested system attains balanced precision, recall, and F1-score values across single-gene, multifactorial, and maternal-mitochondrial disorder categories, as indicated in Table I. The multifactorial disorder category has the highest F1-score of the three classes, demonstrating how well clinical, physiological, and inheritance-based features can be integrated for hereditary risk assessment.

C. Analysis of Confusion Matrix

The confusion matrix for classifying the risk of hereditary disorders is shown in Fig. 4. Limited off-diagonal values show low misclassification rates, while strong diagonal dominance shows a high number of accurate predictions across all disorder categories. This demonstrates that the model can successfully differentiate between various families of hereditary disorder.

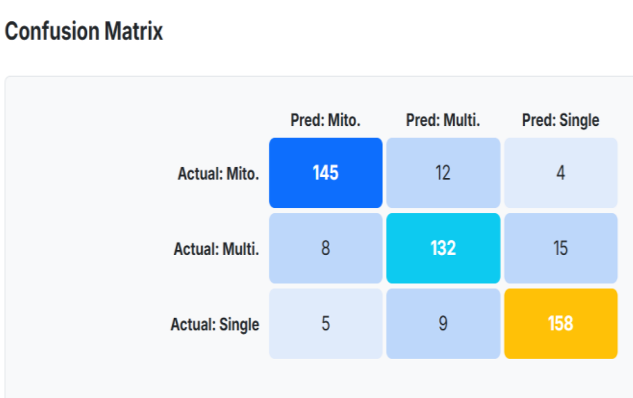

Fig.4.Confusion matrix for hereditary disorder risk classification

D. Predicting Follow-up Urgency

The system successfully prioritized patients for additional clinical screening and consultation by achieving high predictive accuracy for both low-risk and high-risk categories.

A combined analysis of maternal health conditions, pregnancy-related risk factors, family inheritance indicators, and physiological parameters like blood cell counts, heart rate, and respiratory rate is used to determine the urgency classification. Together, these characteristics represent both a genetic predisposition and the state of physiological stress, enabling the model to recognize situations that call for prompt medical intervention. A higher probability of hereditary disorder influence is indicated by high-risk predictions, which also indicate the necessity of advanced diagnostic testing, a specialist referral, or a priority clinical consultation.

E. Feature Importance Analysis

Understanding which features contribute most to the model's decision-making process is essential for improving transparency and clinical relevance. To interpret the behavior of the proposed ensemble model, a feature importance analysis was performed after training.

The results, shown in **Fig. 5**, indicate that maternal genetic history is the most influential factor in predicting hereditary disorder risk. This suggests that inheritance patterns from the maternal side play a significant role in determining susceptibility within the studied dataset. Blood cell count was identified as the second most impactful feature, highlighting the relevance of physiological indicators in risk estimation. Paternal inheritance history also demonstrated a meaningful contribution, reinforcing the importance of family background in hereditary disease assessment.

Patient age, while still relevant, showed comparatively lower influence than inheritance-related and physiological attributes. This finding aligns with the understanding that hereditary disorders are primarily driven by genetic and familial factors, though demographic characteristics can still contribute to overall risk profiling.

The feature importance ranking supports the design choice of integrating both clinical measurements and inheritance indicators.

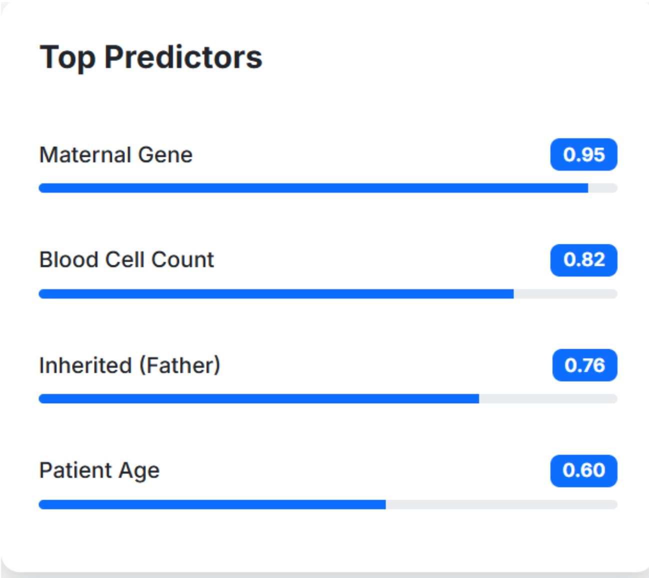

Fig. 5. **Feature importance ranking of input attributes**

E. Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1-Score
Naive Bayes	78.2	0.77	0.75	0.76
Random Forest	86.4	0.86	0.85	0.85
Bagging	88.1	0.88	0.87	0.87
Proposed Ensemble	**90.5**	**0.91**	**0.89**	**0.90**

Table II. Performance comparison of machine learning models

Work	Approach	Uses Genomic / Chromosomal Data	Ensemble ML	Disorder-Level Classification	Disease-Level Diagnosis	Web-Based System	Real-Time Screening
Smith et al. (2020)	Single-gene disease ML model	✓	✗	✗	✓	✗	✗
Kumar et al. (2021)	Genomic deep learning	✓	✗	✗	✓	✗	✗
Zhang et al. (2022)	Rule-based clinical screening	✗	✗	✓	✗	✗	✗
Patel et al. (2023)	ML with chromosomal analysis	✓	✓	✗	✓	✗	✗
Lee et al. (2024)	Ensemble ML on genomic data	✓	✓	✗	✓	✗	✗
Proposed Intelli Gene	Ensemble ML on clinical data	✗	✓	✓	✗	✓	✓

TABLE III. Comparison with Existing Hereditary Disorder Prediction Approaches

V. CONCLUSION

Intelli Gene, a web-based intelligent framework for early hereditary disorder risk assessment using ensemble machine learning techniques, was introduced in this paper. The system provides real-time risk stratification without the need for costly genomic or chromosomal testing by integrating routinely collected clinical, maternal health, familial inheritance, and physiological parameters. The suggested ensemble model outperformed individual classifiers in terms of accuracy, robustness, and generalization by combining Random Forest, Bagging, and Naive Bayes classifiers. The experimental results showed good follow-up urgency prediction and consistent performance across single-gene, multifactorial, and maternal-mitochondrial disorder categories. Practical usability for clinical decision support and early screening is made possible by the scalable web deployment. To further improve clinical adoption and predictive reliability, future research may concentrate on integrating explainable AI techniques, larger clinical datasets, and electronic health record systems

VI. FUTURE WORK

Using structured clinical and inheritance-based data, the suggested Intelli Gene framework shows good performance for early-stage hereditary disorder risk screening. To further increase the system's functionality and clinical relevance, a number of improvements could be investigated in subsequent research. Beyond risk classification, more accurate disorder-level diagnosis may be possible through the integration of genomic and chromosomal data, such as single nucleotide polymorphisms (SNPs) and karyotype information. Predictive accuracy may be improved and disease progression supported by the incorporation of longitudinal patient records and temporal health trends. In order to enhance model interpretability and transparency for clinical decision support, future extensions might also incorporate explainable artificial intelligence (XAI) techniques. Multi-institutional partnerships that increase the size of the dataset can improve generalization and lessen data bias. Furthermore, large-scale deployment and automated screening workflows can be made easier by real-time integration with hospital information systems and electronic health records (EHRs). The adoption of advanced ensemble optimization and deep learning models may further enhance predictive robustness while maintaining ethical and regulatory compliance.

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